Global Energy Consumption Visualization Dashboard

MINI PROJECT REPORT

Submitted by

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BONAFIDE CERTIFICATE

*This is to Certified that this Mini Project report "**GLOBAL ENERGY CONSUMPTION VISUALIZATION DASHBOARD**" is the Bonafide work of **"DEVIHARINI M (2117240030026)**"* who carried out the project work under my supervision.

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INTERNAL EXAMINER ***EXTERNAL EXAMINER***

INTRODUCTION

A **Global Energy Consumption Visualization Dashboard** is an interactive digital platform designed to present complex energy data in a clear, intuitive, and visually engaging way. As the world faces growing challenges related to climate change, resource depletion, and sustainable development, understanding how energy is produced, distributed, and consumed across regions becomes increasingly important. This dashboard serves as a powerful tool for policymakers, researchers, businesses, and the general public to explore global energy trends and make informed decisions.

By integrating data from reliable sources such as the International Energy Agency and the World Bank, the dashboard provides insights into various energy sources, including fossil fuels, renewables, nuclear, and emerging technologies. Users can analyze consumption patterns by country, region, or sector, track historical changes, and compare energy usage across different economies.
Advanced visualization techniques—such as interactive maps, time-series graphs, and dynamic charts—allow users to quickly identify key patterns, trends, and anomalies. For instance, one can observe the rapid growth of renewable energy adoption or examine disparities in energy access between developed and developing regions.

Ultimately, the Global Energy Consumption Visualization Dashboard aims to bridge the gap between raw data and actionable knowledge. By transforming large datasets into meaningful visual stories, it supports global efforts toward energy efficiency, sustainability, and a more resilient energy future.

PROBLEM STATEMENT

The rapid growth in global energy demand, coupled with the urgent need to transition toward sustainable energy systems, has made it increasingly difficult to interpret and utilize large volumes of energy-related data effectively. Data from organizations such as the International Energy Agency and the World Bank is often vast, fragmented, and presented in formats that are not easily accessible or understandable to all stakeholders.

Currently, there is a lack of a unified, user-friendly platform that consolidates global energy consumption data and presents it through interactive and meaningful visualizations. Policymakers, researchers, and analysts face challenges in identifying consumption patterns, comparing energy usage across countries, and tracking the progress of renewable energy adoption due to scattered datasets and limited visualization tools.

This problem is further compounded by the need for real-time or near-real-time insights, as well as the ability to customize views based on specific regions, sectors, or energy sources. Without an efficient visualization system, critical decision-making related to energy planning, sustainability goals, and climate action is hindered.

Therefore, there is a need to develop a comprehensive Global Energy Consumption Visualization Dashboard that transforms complex datasets into clear, interactive visual insights, enabling users to better understand global energy trends and support data-driven decision-making.

PROJECT GOAL

The primary goal of the Global Energy Consumption Visualization Dashboard is to design and develop an interactive, intuitive platform that transforms complex global energy data into clear, accessible, and actionable visual insights. By leveraging reliable datasets from organizations such as the International Energy Agency and the World Bank, the project aims to provide a centralized solution for exploring and analyzing energy consumption patterns worldwide.

The dashboard is intended to enable users to monitor and compare energy usage across different countries, regions, and sectors, while also tracking trends over time. A key goal is to highlight the shift toward renewable energy sources and support awareness of sustainability challenges, helping stakeholders better understand the global energy landscape.

In addition, the project seeks to enhance decision-making by offering interactive features such as dynamic filtering, real-time updates, and customizable visualizations. These capabilities will allow policymakers, researchers, and businesses to derive meaningful insights, identify inefficiencies, and evaluate the impact of energy policies and initiatives.

Ultimately, the goal is to bridge the gap between raw energy data and informed action, contributing to global efforts in energy efficiency, climate responsibility, and long-term sustainable development.

THEORETICAL BACKGROUND

Theoretical background refers to the fundamental concepts and principles that support the development of this project. The Global Energy Consumption Visualization Dashboard is based on concepts from data analysis, energy studies, statistics, and data visualization. Understanding these theoretical concepts helps in analyzing global energy usage patterns and presenting meaningful insights through visual representations.

1. Energy Consumption
Energy consumption refers to the total amount of energy used by individuals, industries, and countries over a specific period of time. It includes different energy sources such as fossil fuels (coal, oil, natural gas), nuclear energy, and renewable energy (solar, wind, hydro). Monitoring energy consumption is important for understanding resource usage, environmental impact, and sustainability.

2. Global Energy Data
Global energy data consists of information collected from different countries regarding their energy production and consumption. This data includes variables such as energy type, consumption levels, population, and time period. Analyzing this data helps identify how energy usage varies across regions and over time.

3. Time Series Data
Time series data refers to a sequence of data points collected at regular intervals over time. In this project, energy consumption data is analyzed over multiple years. This helps in identifying trends such as increasing or decreasing energy usage and understanding long-term patterns in global energy consumption.

4. Data Analysis
Data analysis involves inspecting, cleaning, and transforming raw data to extract useful information. In the context of this project, data analysis is used to compare energy consumption across countries, identify high and low consumption regions, and detect patterns or anomalies in the dataset.

5. Data Visualization
Data visualization is the graphical representation of data using charts, graphs, and maps. In this dashboard, visualization techniques such as line charts, bar graphs, and pie charts are used to present energy consumption data. Visualization makes complex data easier to understand and helps users quickly identify trends and patterns.

6. Dashboard Systems
A dashboard is an interactive interface that displays key information in a visual format. It allows users to explore data dynamically by applying filters and selecting different parameters. The dashboard in this project provides a user-friendly way to view and analyze global energy consumption data efficiently.

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7. Web-Based Technologies
Modern dashboards are often built using web technologies such as HTML, CSS, and JavaScript. These technologies help create responsive and interactive user interfaces. Libraries like charts and visualization tools enable real-time data display and improve the overall user experience.

ALGORITHM EXPLANATION

Step 1: Data Collection
Collect global energy consumption data from reliable sources such as datasets or CSV files. The data typically includes attributes like country name, year, type of energy source, and consumption values.

Step 2: Data Preprocessing
Clean and prepare the dataset by handling missing values, removing duplicate entries, and converting data into a suitable format. This ensures accuracy and consistency in analysis.

Step 3: Data Loading
Load the processed dataset into the system or web application. This can be done using programming languages like JavaScript (for web apps) or Python (for backend processing).

Step 4: Data Filtering
Allow users to filter data based on parameters such as country, year, or energy type. This helps in focusing on specific subsets of data for better analysis.

Step 5: Data Analysis

Perform basic analysis to identify trends and patterns. For example, calculate total energy consumption, compare consumption between countries, or observe changes over time.

Step 6: Visualization Generation

Generate visual representations of the data using charts such as:

Line charts (for trends over time)

Bar charts (for comparison between countries)

Pie charts (for energy source distribution)

These visuals make the data easier to understand.

Step 7: Dashboard Display

Display the visualizations on a dashboard interface. The dashboard updates dynamically based on user inputs and selected filters.

Step 8: User Interaction

Enable user interaction such as selecting options, hovering over charts, or switching between different views. This improves usability and helps users explore the data effectively.

Step 9: Result Output

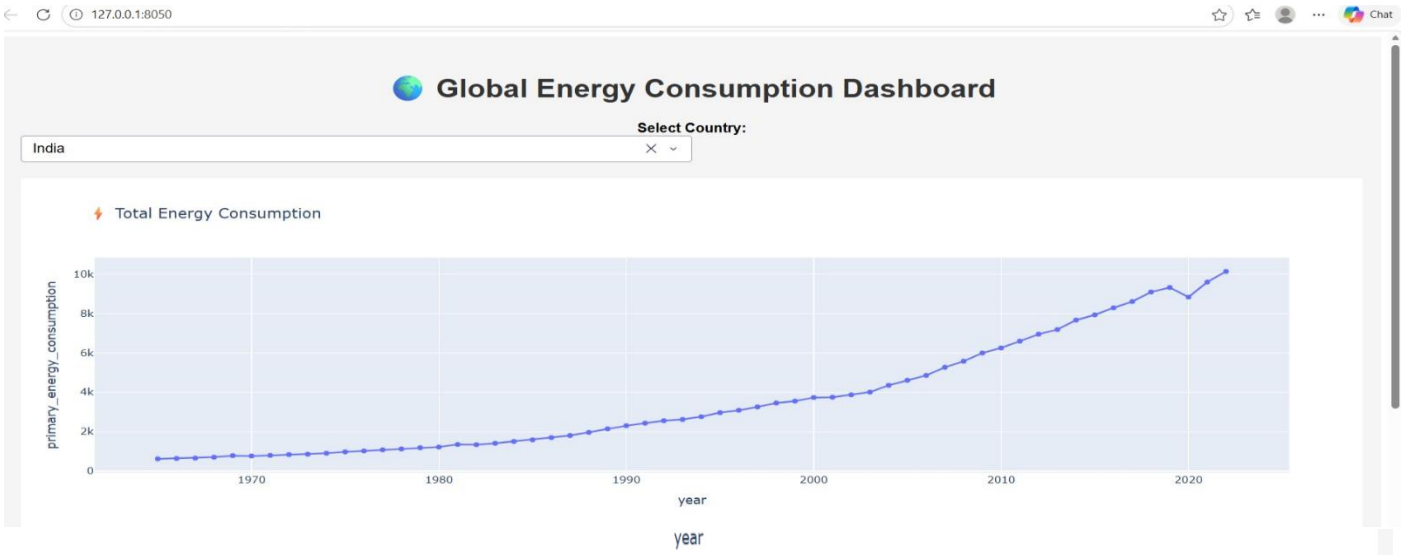
Present the final analyzed data and visual insights to the user in an easy-to-understand format, helping in decision-making and interpretation.

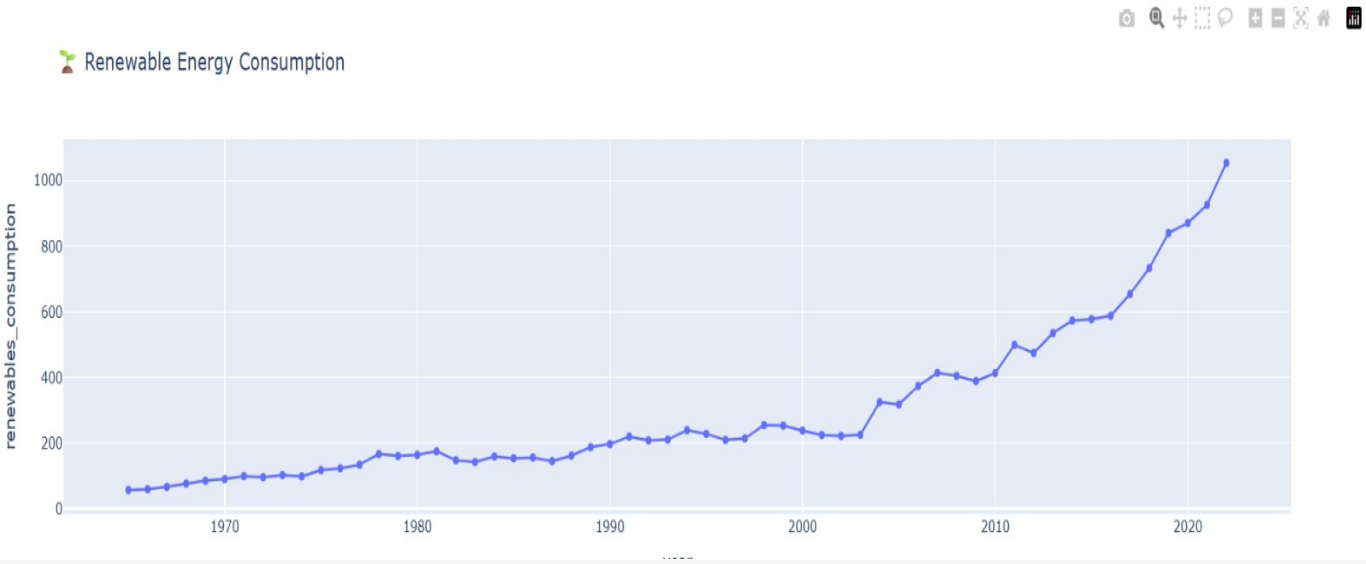
IMPLEMENTATION AND CODE

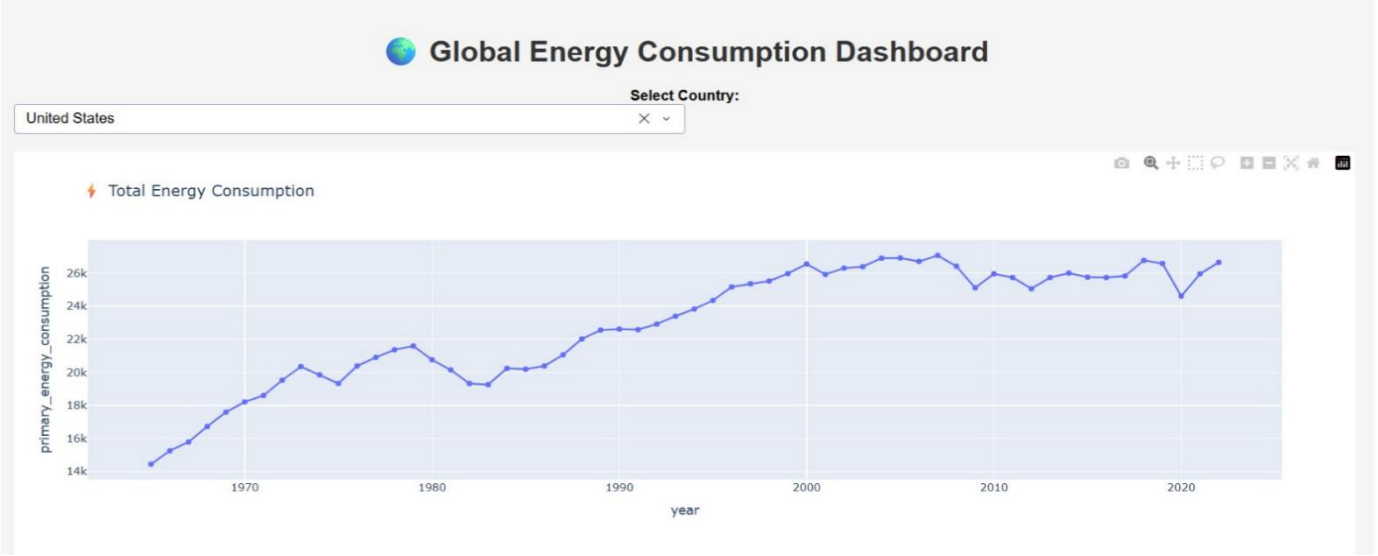
```
 import pandas as pd
from dash import Dash, dcc, html, Input, Output
import plotly.express as px

# Load dataset
df = pd.read_csv("energy.csv")

# Create app
app = Dash(__name__)

# Layout
app.layout = html.Div([
```

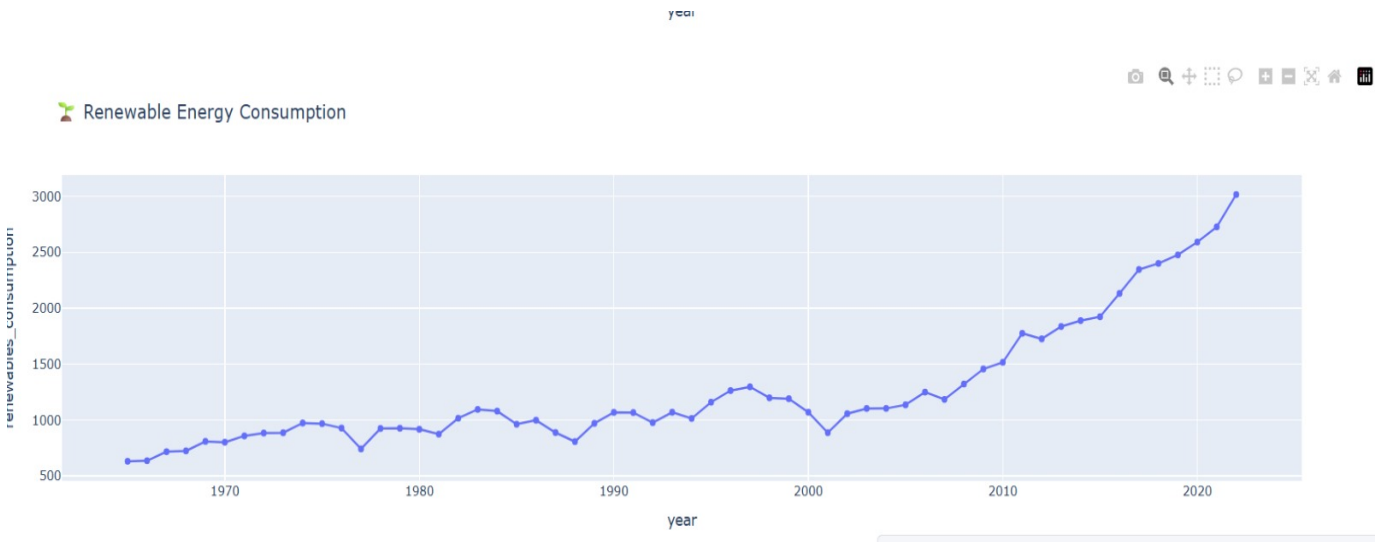
```
    html.H1("Global Energy Consumption Dashboard", style={'textAlign': 'center'}),

    # Dropdown
    dcc.Dropdown(
        id='country-dropdown',
        options=[{'label': c, 'value': c} for c in df['country'].dropna().unique()],
        value='India'
    ),

    # Graphs
    dcc.Graph(id='energy-graph'),
    dcc.Graph(id='renewable-graph')
])

# Callback
@app.callback(
    [Output('energy-graph', 'figure'),
     Output('renewable-graph', 'figure')],
    [Input('country-dropdown', 'value')]
)

def update_graphs(selected_country):
    data = df[df['country'] == selected_country]

    # Energy graph
    fig1 = px.line(
        data, x='year', y='primary_energy_consumption',
        title='Energy Consumption Over Time'
    )

    # Renewable graph
    fig2 = px.line(
        data, x='year', y='renewable_energy_share',
        title='Renewable Energy Share'
    )

    return fig1, fig2

# Run app
if __name__ == '__main__':
    app.run(debug=True)
```

OUTPUT


127.0.0.1:8050
Chat
Global Energy Consumption Dashboard
Select Country:
India
Total Energy Consumption
primary_energy_consumption
10k
8k
6k
4k
2k
0
1970
1980
1990
2000
2010
2020
year


Renewable Energy Consumption
renewables_consumption
1000
800
600
400
200
0
1970
1980
1990
2000
2010
2020


Global Energy Consumption Dashboard
Select Country:
United States
Total Energy Consumption
primary_energy_consumption
26k
24k
22k
20k
18k
16k
14k
1970
1980
1990
2000
2010
2020
year



DATASET FOR ANALYSIS:

data.csv

```
country,year,energy,renewable
India,2018,600,20
India,2019,650,25
India,2020,700,30
USA,2018,1200,15
USA,2019,1250,18
USA,2020,1300,20
China,2018,1500,10
China,2019,1600,12
China,2020,1700,15
```

RESULTS AND FUTURE ENHANCEMENT

Results: The Global Energy Consumption Visualization Dashboard successfully presents energy usage data in a clear and interactive manner. The system enables users to analyze global energy consumption patterns through various visual tools and filters.

The dashboard displays energy consumption data for different countries and time periods using charts such as line graphs, bar charts, and pie charts. These

visualizations help in easily identifying trends, comparisons, and distributions of energy usage.

From the analysis, it is observed that:

- Energy consumption varies significantly across countries depending on population and industrial development.
- Fossil fuels remain the dominant source of energy in many regions, while renewable energy usage is gradually increasing.
- Over the years, there is a noticeable rise in global energy consumption due to economic growth and technological advancements.
- Some countries show a shift towards cleaner and sustainable energy sources.

The filtering feature allows users to focus on specific countries, years, or energy types, making the analysis more flexible and user-friendly. The interactive dashboard improves understanding by providing real-time updates based on user input.

Overall, the system effectively transforms complex energy data into meaningful visual insights, helping users understand global energy trends and supporting better decision-making.

Future Enhancements:

The current system provides basic visualization and analysis of global energy consumption data. However, several improvements can be made to enhance its functionality and usability in the future.

- **Real-Time Data Integration**
  The current system provides basic visualization and analysis of global energy consumption data. However, several improvements can be made to enhance its functionality and usability in the future.
- **Real-Time Data Integration**
  The dashboard can be enhanced by integrating real-time energy data using APIs. This will allow users to view the most recent energy consumption statistics instead of relying on static datasets.
- **Advanced Analytics and Forecasting**
  Future versions can include predictive analysis using machine learning techniques to forecast energy consumption trends. This will help in planning and decision-making.
- **Geographical Visualization (Map Integration)**
  Adding interactive world maps can help users visualize energy consumption geographically. This makes it easier to compare energy usage across different regions.

- **User Authentication and Personalization**
  Implementing user login features will allow users to save preferences, customize dashboards, and track specific countries or energy types.
- **Mobile-Friendly Interface**
  Optimizing the dashboard for mobile devices will improve accessibility and allow users to access the system from smartphones and tablets.
- **Detailed Energy Source Analysis**
  The system can be expanded to provide deeper insights into individual energy sources such as solar, wind, and hydro energy.
- **Export and Reporting Features**
  Users can be given options to download reports in formats like PDF or Excel for further analysis and documentation.
- **Performance Optimization**
  Improving the system's performance for handling large datasets will ensure faster data processing and smoother user experience.

Git Hub Link for the project and report

Github link Of the Project and Report	

REFERENCES

1 International Energy Agency (IEA), *World Energy Outlook Reports* https://www.iea.org/

2 U.S. Energy Information Administration (EIA), *International Energy Data* https://www.eia.gov/

3 World Bank, *Energy Use and Statistics Database* https://data.worldbank.org/

4 BP, *Statistical Review of World Energy*
https://www.bp.com/

5 Our World in Data, *Energy Consumption Dataset*
ttps://ourworldindata.org/

6 Introduction to Data Mining, Pearson Education

7 Python for Data Analysis, O'Reilly Media

8 Microsoft Excel – for data preprocessing and analysis

9 Tableau / Power BI – for dashboard creation

10 Visual Studio Code – for development